DBMS Teaching Material

AI Virtual Teaching Assistant – Sample Knowledge Base

1. Introduction to DBMS

A Database Management System (DBMS) is software used to create, store, organize, retrieve, update, and manage data efficiently. Examples include MySQL, PostgreSQL, Oracle Database, and Microsoft SQL Server.

Advantages: reduced data redundancy, improved consistency, security, data sharing, backup and recovery, and controlled concurrent access.

2. Database Models

Common database models include hierarchical, network, relational, object-oriented, and NoSQL models. The relational model represents information using tables consisting of rows (tuples) and columns (attributes).

3. Keys in DBMS

Primary Key: uniquely identifies each row in a table and cannot contain NULL values.
Candidate Key: a minimal attribute set that can uniquely identify a row.
Foreign Key: an attribute that references a key in another table.
Composite Key: a key made from two or more attributes.

4. ER Model

The Entity-Relationship (ER) model represents entities, attributes, and relationships. An entity is a distinguishable object, an attribute describes an entity, and a relationship shows an association between entities. Cardinality may be one-to-one, one-to-many, many-to-one, or many-to-many.

5. SQL Basics

SQL (Structured Query Language) is used to communicate with relational databases. DDL includes CREATE, ALTER, and DROP. DML includes INSERT, UPDATE, and DELETE. DQL commonly uses SELECT. DCL includes GRANT and REVOKE, while TCL includes COMMIT and ROLLBACK.

Example: SELECT name, marks FROM Student WHERE marks > 70;

6. Normalization

Normalization organizes tables to reduce redundancy and avoid update anomalies. **1NF** requires atomic values and no repeating groups. **2NF** requires 1NF and removes partial dependency on a composite key. **3NF** requires 2NF and removes transitive dependency. **BCNF** is a stronger form of 3NF in which every determinant is a candidate key.

7. Transactions and ACID Properties

A transaction is a logical unit of database work. Its common states include active, partially committed, committed, failed, and aborted. **ACID** means Atomicity, Consistency, Isolation, and Durability.

8. Concurrency Control

Concurrency control ensures that simultaneous transactions do not produce incorrect results. Common techniques include locking, timestamp ordering, and optimistic methods. Locks may be shared (read) or exclusive (write).

9. Deadlock

A deadlock occurs when transactions wait indefinitely for resources held by one another. The four classic necessary conditions are mutual exclusion, hold and wait, no preemption, and circular wait. Deadlock can be handled using prevention, avoidance, detection, and recovery.

10. Important Questions for Students

1. What is a DBMS? Explain its advantages.
2. Explain primary, candidate, and foreign keys with examples.
3. Explain the ER model and cardinality.
4. Differentiate DDL, DML, DCL, DQL, and TCL.
5. Explain 1NF, 2NF, 3NF, and BCNF with examples.
6. Explain ACID properties and transaction states.
7. Explain concurrency control and locking techniques.
8. What is deadlock? Explain its necessary conditions and handling methods.

End of Teaching Material